CAPTURE THE FLAG

WRITEUP

RedLine Lab

cyberdefenders.org • Memory Forensics

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Alias	CYB3R_B0T
Category	Memory Forensics
Difficulty	Easy
Date Solved	15-09-2026
Status	Solved

1 Executive Summary

The memory forensics investigation was performed on a hacked Windows machine through Volatility 3. The investigation centered on the identification of any suspicious processes, network connections, malware memory objects, and possible C2 communication. The investigation revealed IoCs that confirm the existence of malware infection within the computer system.

2 Challenge Information

Challenge Name	Redline Lab
Category	Memory Forensics
Points	7/7
Difficulty	Easy
Files Provided	MemoryDump.mem
Target	Find answers to several questions through memory dump analysis.
Author	Md Solayman Hossain Shawon

2.1 Challenge Description

This challenge focuses on analyzing a Windows memory dump using Volatility 3 to investigate suspicious processes, network activity, and potential malicious artifacts. The objective is to identify key indicators of compromise and answer several forensic investigation questions.

2.2 Objective

Analyze the provided memory dump using Volatility 3 to identify suspicious activity, malicious processes, network connections, and key indicators of compromise.

3 Reconnaissance & Initial Analysis

Document everything discovered before an attack path was chosen. This grounds the writeup and lets a reader retrace your reasoning.

3.1 Environment / File Triage

- The provided forensic evidence has been analyzed for any suspicious activities including processes, files, network connections, and system activity.
- Process and Network information was checked to see any malicious or suspicious activities.
- Process name, PID, File path, and Network Connections were correlated to find any suspicious activity.
- Examined command-line activities and found an unusual PHP command, which needed further inspection.

3.2 Observations

- Suspect executable: oneetx.exe was found executing from a suspected temporary user directory.
- Suspect PHP command: A PHP command was found executed via the command line and was classified as suspicious because of the nature of the execution.
- Network activity: Suspect external communications were detected from processes such as oneetx.exe and tun2socks.exe.
- VPN-related activity: Network communication and process information were correlated in order to determine the process performing the VPN activity.
- Indicators of malicious activity: The combination of a suspect location of the executable, execution of the PHP command and suspect network activity required further analysis.

4 Methodology / Approach

Summarize the strategy chosen and why, before walking through execution. This is the “how I thought about it” section.

4.1 Tools Used

Tool	Purpose
Volatility 3	RAM Memory Analysis
Select-String	Filtering certain keywords/patterns from output of command or text files

4.2 Hypothesis

The system might have been hacked due to the suspicious execution of commands, followed by the execution of a malicious executable file and further networking.

5 Step-by-Step Walkthrough

Walk through the solve path in the order it happened, with enough commands and screenshots that another player could reproduce it exactly.

Question 1 — What is the name of the suspicious process?

Step 1 – Open Powershell and Enumerate Running Processes Using Volatility 3.

```
python vol.py -f  C:\Users\User\Documents\volatility-web\memory\MemoryDump.mem windows.pslist
```

Step 2 – After analyzing all entries under **ImageFileName** , **oneetx.exe** was flagged as an unusual process that warrants additional analysis.

Answer: oneetx.exe

Question 2 – What is the child process name of the suspicious process?

Step 1 – Based on the previous findings, oneetx.exe was identified as the suspicious executable. Its corresponding **Process ID (PID)** was obtained from the process listing.

Step 2 – Next, we enumerated the running process tree to establish the parent-child relationship of the suspicious process using the following command:

```
python vol.py -f  C:\Users\User\Documents\volatility-web\memory\MemoryDump.mem windows.pstree --pid 5896
```

Step 3 – After analyzing the process tree, we found that **rundll32.exe** had **PPID 5896**, which corresponds to oneetx.exe. Therefore, rundll32.exe was identified as the **child process** of the suspicious process.

Answer: rundll32.exe

Question 3 — What is the memory protection applied to the suspicious process memory region?

Step 1 – From the previous analysis, oneetx.exe was identified as the suspicious executable with **PID 5896**.

Step 2 – We examined the Virtual Address Descriptors (VADs) of PID 5896 using:

```
python vol.py -f  C:\Users\User\Documents\volatility-web\memory\MemoryDump.mem windows.vadinfo --pid5896
```

Step 3 – The VAD output contained several memory regions. We focused on regions with:

PAGE_EXECUTE_READWRITE

This permission allows a memory region to be **read, written, and executed**, making it an important indicator for further malware analysis.

Answer: PAGE_EXECUTE_READWRITE

Question 4 — What is the name of the process responsible for the VPN connection?

Step 1 – We analyzed the active and historical network connections from the memory dump using Volatility 3's windows.netscan plugin:

```
python vol.py -f  C:\Users\User\Documents\volatility-web\memory\MemoryDump.mem windows.netscan
```

Step 2 – After reviewing the **windows.netscan** output, we initially identified tun2socks.exe as a potential candidate because it was associated with network activity. However, further analysis showed that **tun2socks.exe** was not the actual process responsible for the VPN connection.

Step 3 – From the previous analysis, PID **6724** was identified as the relevant PPID. We then analyzed the process list using the **windows.pslist** plugin:

```
python vol.py -f C:\Users\User\Documents\volatility-web\memory\MemoryDump.mem windows.pslist
```

After searching for PID **6724**, we found that its ImageFileName was **Outline.exe.** This identified **Outline.exe** as the process associated with the VPN connection.

Answer: **Outline.exe**

Question 5 — What is the attacker's IP address?

Step 1 – As seen in the above discussion, PID 5896 turned out to be the PID of the suspicious executable named oneetx.exe. We then proceeded to analyze the network connections for this PID.

The command we used to filter out the results from windows.netscan with PID 5896 is as follows:

```
python vol.py -f C:\Users\User\Documents\volatility-web\memory\MemoryDump.mem windows.netscan | Select-String "5896"
```

Step 2 – After analyzing the filtered netscan output, we identified that oneetx.exe (PID **5896**) had a network connection to the remote IP address **77.91.124.20**.

Answer: 77.91.124.20

Question 6 — What is the full URL of the PHP file that the attacker visited?

Step 1 – From the previous analysis, 77.91.124.20 was identified as the remote IP associated with the suspicious activity. We searched the memory dump for this IP address to identify any related URLs.

The command we used to filter out the results:

```
Select-String -Path"C:\Users\User\Documents\volatility-web\memory\MemoryDump.mem" -Pattern "77.91.124.20"
```

Step 2 – The search results revealed a reference to the following PHP file:

Answer: http://77.91.124.20/store/games/js/index.php

Question 7 —What is the full path of the malicious executable?

Step 1 – From the previous analysis, oneetx.exe was identified as the malicious executable with **PID 5896**.

Step 2 – We analyzed the process tree for PID 5896:

```
python vol.py -f  C:\Users\User\Documents\volatility-web\memory\MemoryDump.mem windows.pstree --pid 5896
```

This confirmed the process and its parent-child relationship.

Step 3 - Now we found C:\Users\Tammam\AppData\Local\Temp\c3912af058\oneetx.exe

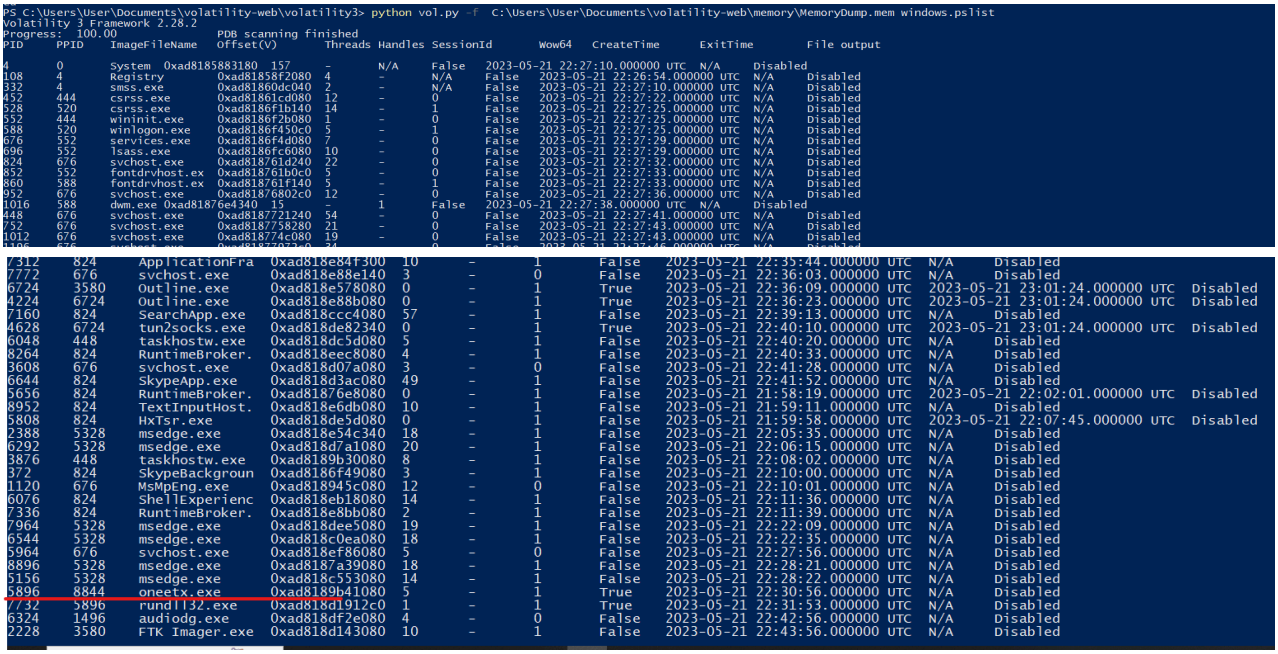
Answer: C:\Users\Tammam\AppData\Local\Temp\c3912af058\oneetx.exe

6 Proof of Concept

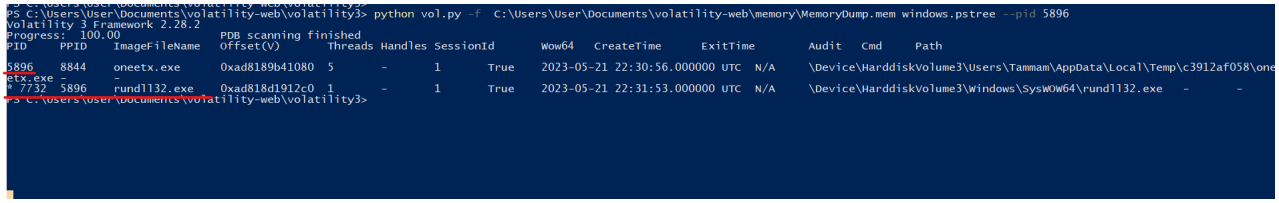
Provide the final, cleaned-up exploit so it can be run end-to-end and independently verified.

6.1 Execution Output

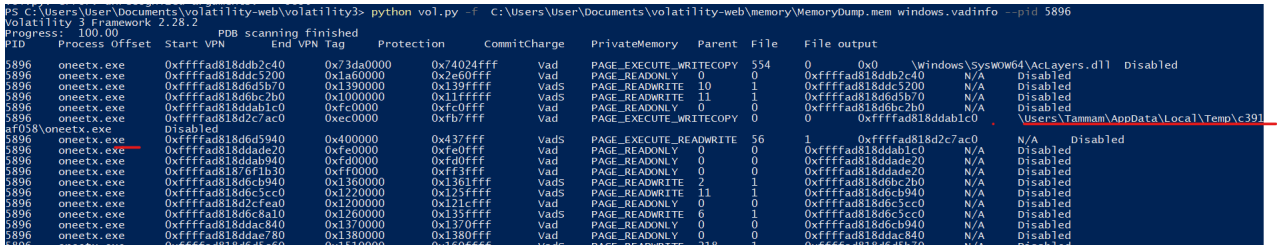
Questions 1:



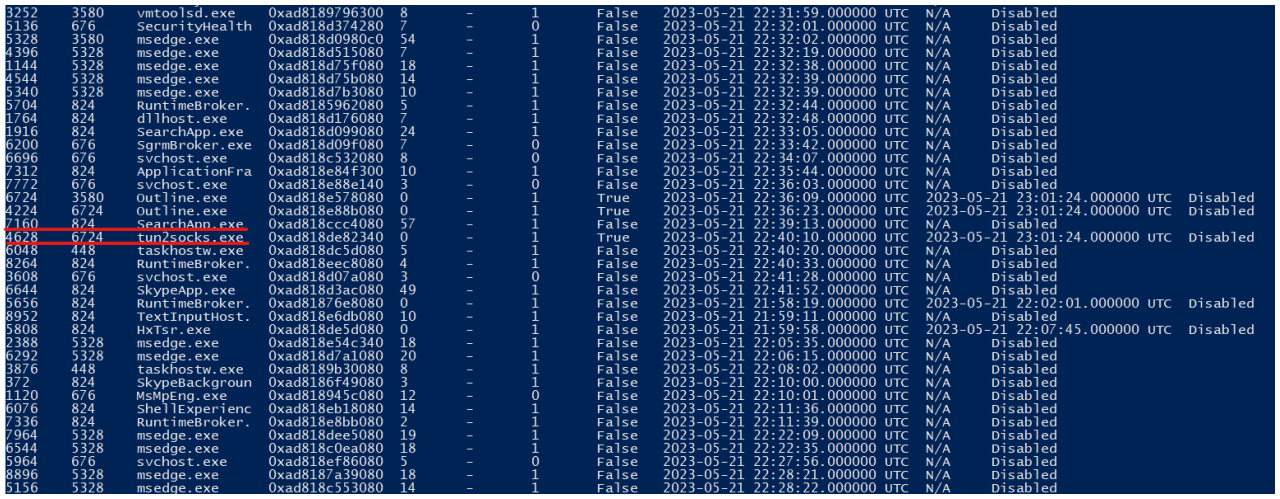
Questions 2:



Questions 3:

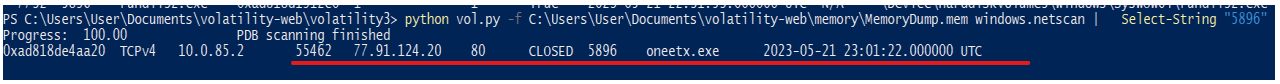


Questions 4:



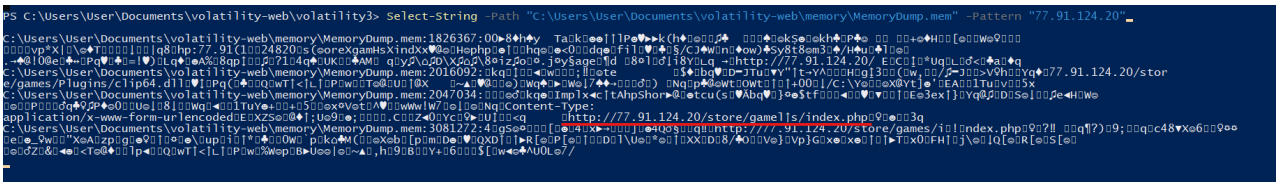
```
3252    3580    vmtoolsd.exe    0xad8189796300  8       -       1       False   2023-05-21 22:31:59.000000 UTC  N/A     Disabled
5136    676     SecurityHealth  0xad818d374280  7       -       0       False   2023-05-21 22:32:01.000000 UTC  N/A     Disabled
5328    3580    msedge.exe      0xad818d0980c0  54      -       1       False   2023-05-21 22:32:02.000000 UTC  N/A     Disabled
4396    5328    msedge.exe      0xad818d515080  7       -       1       False   2023-05-21 22:32:19.000000 UTC  N/A     Disabled
1144    5328    msedge.exe      0xad818d75f080  18      -       1       False   2023-05-21 22:32:38.000000 UTC  N/A     Disabled
4544    5328    msedge.exe      0xad818d75b080  14      -       1       False   2023-05-21 22:32:39.000000 UTC  N/A     Disabled
5340    5328    msedge.exe      0xad818d7b3080  10      -       1       False   2023-05-21 22:32:39.000000 UTC  N/A     Disabled
5704    824     RuntimeBroker.  0xad8185962080  5       -       1       False   2023-05-21 22:32:44.000000 UTC  N/A     Disabled
1764    824     dllhost.exe     0xad818d176080  7       -       1       False   2023-05-21 22:32:48.000000 UTC  N/A     Disabled
1916    824     SearchApp.exe   0xad818d099080  24      -       1       False   2023-05-21 22:33:05.000000 UTC  N/A     Disabled
6200    676     SgrmBroker.exe  0xad818d09f080  7       -       0       False   2023-05-21 22:33:42.000000 UTC  N/A     Disabled
6696    676     svchost.exe     0xad818c532080  8       -       0       False   2023-05-21 22:34:07.000000 UTC  N/A     Disabled
7312    824     ApplicationFra  0xad818e84f300  10      -       1       False   2023-05-21 22:35:44.000000 UTC  N/A     Disabled
7772    676     svchost.exe     0xad818e88e140  3       -       0       False   2023-05-21 22:36:03.000000 UTC  N/A     Disabled
6724    3580    Outline.exe     0xad818e578080  0       -       1       True    2023-05-21 22:36:09.000000 UTC  2023-05-21 23:01:24.000000 UTC  Disabled
4224    6724    Outline.exe     0xad818e88b080  0       -       1       True    2023-05-21 22:36:23.000000 UTC  2023-05-21 23:01:24.000000 UTC  Disabled
7160    824     SearchApp.exe   0xad818ccc4080  57      -       1       False   2023-05-21 22:39:13.000000 UTC  N/A     Disabled
4628    6724    tun2socks.exe   0xad818de82340  0       -       1       True    2023-05-21 22:40:10.000000 UTC  2023-05-21 23:01:24.000000 UTC  Disabled
6048    448     taskhostw.exe   0xad818dc5d080  5       -       1       False   2023-05-21 22:40:20.000000 UTC  N/A     Disabled
8264    824     RuntimeBroker.  0xad818eec8080  4       -       1       False   2023-05-21 22:40:33.000000 UTC  N/A     Disabled
3608    676     svchost.exe     0xad818d07a080  3       -       0       False   2023-05-21 22:41:28.000000 UTC  N/A     Disabled
6644    824     SkypeApp.exe    0xad818d3ac080  49      -       1       False   2023-05-21 22:41:52.000000 UTC  N/A     Disabled
5656    824     RuntimeBroker.  0xad81876e8080  0       -       1       False   2023-05-21 21:58:19.000000 UTC  2023-05-21 22:02:01.000000 UTC  Disabled
8952    824     TextInputHost.  0xad818e6db080  10      -       1       False   2023-05-21 21:59:11.000000 UTC  N/A     Disabled
5808    824     HxTsr.exe       0xad818de5d080  0       -       1       False   2023-05-21 21:59:58.000000 UTC  2023-05-21 22:07:45.000000 UTC  Disabled
2388    5328    msedge.exe      0xad818e54c340  18      -       1       False   2023-05-21 22:05:35.000000 UTC  N/A     Disabled
6292    5328    msedge.exe      0xad818d7a1080  20      -       1       False   2023-05-21 22:06:15.000000 UTC  N/A     Disabled
3876    448     taskhostw.exe   0xad8189b30080  8       -       1       False   2023-05-21 22:08:02.000000 UTC  N/A     Disabled
372     824     SkypeBackgroun  0xad8186f49080  3       -       1       False   2023-05-21 22:10:00.000000 UTC  N/A     Disabled
1120    676     MsMpEng.exe     0xad818945c080  12      -       0       False   2023-05-21 22:10:01.000000 UTC  N/A     Disabled
6076    824     ShellExperienc  0xad818eb18080  14      -       1       False   2023-05-21 22:11:36.000000 UTC  N/A     Disabled
7336    824     RuntimeBroker.  0xad818e8bb080  2       -       1       False   2023-05-21 22:11:39.000000 UTC  N/A     Disabled
7964    5328    msedge.exe      0xad818dee5080  19      -       1       False   2023-05-21 22:22:09.000000 UTC  N/A     Disabled
6544    5328    msedge.exe      0xad818c0ea080  18      -       1       False   2023-05-21 22:22:35.000000 UTC  N/A     Disabled
5964    676     svchost.exe     0xad818ef86080  5       -       0       False   2023-05-21 22:27:56.000000 UTC  N/A     Disabled
8896    5328    msedge.exe      0xad8187a39080  18      -       1       False   2023-05-21 22:28:21.000000 UTC  N/A     Disabled
5156    5328    msedge.exe      0xad818c553080  14      -       1       False   2023-05-21 22:28:22.000000 UTC  N/A     Disabled
```

Questions 5:



```
PS C:\Users\User\Documents\volatility-web\volatility3> python vol.py -f C:\Users\User\Documents\volatility-web\memory\MemoryDump.mem windows.netscan |   Select-String "5896"
Progress:  100.00               PDB scanning finished
0xad818de4aa20  TCPv4    10.0.85.2       55462   77.91.124.20    80      CLOSED  5896    oneetx.exe      2023-05-21 23:01:22.000000 UTC
```

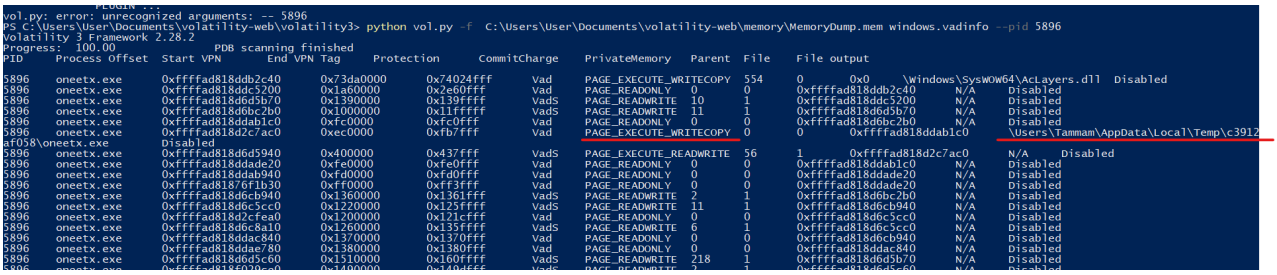
Questions 6:



```
PS C:\Users\User\Documents\volatility-web\volatility3> Select-String -Path "C:\Users\User\Documents\volatility-web\memory\MemoryDump.mem" -Pattern "77.91.124.20"

C:\Users\User\Documents\volatility-web\memory\MemoryDump.mem:1826367:OO▶8♦h♦y  Ta□k□ee↑|lPe♥▶▶k(h♦□e□□♪♦  □□□♦□e◆k$e□◆kh♦□P♦▶ □□ □□+e◆H□□[e□□We♀□□□
□□□□vp*X|□\◆T□□□□□q8□hp:77.91(1□□24820□□(eoreXqamHsXindXx♥◘e□Hsphp□e□□□hqe□e<O□□dqe♦fi□□♥□♦□§/CJ◆W□n□◆ow)◆Sy8t8◆m3□◆/H◆u□◆□De□
.→◆0!0@e□+□Pq♥□◆□=!♥)□Lq◆□e◆X□8qp□□□p□?1□4q◆UK□□◆AM□ q□y□√□o2D\X□o2\8◆iz□o□◆.j◆y$age□¶d □8◆1□d□i8¥□Lq →□http://77.91.124.20/ E□C□I□□*Uq□L□c<□◆a□◆q
C:\Users\User\Documents\volatility-web\memory\MemoryDump.mem:2016092:□kq□I□□◄□w□□□;I□qte      □\◆bq♥□D~JTu□▼Y"!t~yA□□□HGq□3□□(□w,□□/□▬J□□□>v9h□□Yq◆□77.91.124.20/stor
e/games/Plugins/clip64.dll□♥I□Pq(□◆□□□□wTI<lL□P□w□□T◘@□U□T@x   □~▲□♥□□□e)□Wq◆□▶□We□7◆◆→□□□□d□) □Nq□p◆□GwTt□OWt□I□I+00□I/C:\Ye□□eX@YtJe'□EA□□1Tu□v□□5x
C:\Users\User\Documents\volatility-web\memory\MemoryDump.mem:2047034:□□□e□□kqe□Imp1x◄c|tAhpShor►@□etcu(s□♥Abq♥□}♦e$tf□□□◄□□♥□▼□□!□E◘3ex|}□Yq□2□□D□S◘|□□Je◄H□We
□◘□□P□□□□d◆♥□P◆◘O□□Us|□8|□□Wq|◄□□1TuY◘+□□+□5□□e◘V◘t□A♥□□wwwIW7□◘|□◘□Nq□Content-Type:
application/x-www-form-urlencoded□E□XZS◘@□◆|;Ue9E◘;□□□□.C□□z◄0□Vc□9►□U|□□<q      □http://77.91.124.20/store/gamelis/index.php□9□e□□3q
C:\Users\User\Documents\volatility-web\memory\MemoryDump.mem:3081272:4□gSo◘□□□[□◘□4□x►-□□□|□◆4□q$□□q□□http://77.9L.124.20/store/games/i□I□ndex.php□9□?‼ □□q¶?)□9;□□q□c48▼Xø6□□9◘◘
□e□e_9w□□"XøA□zp□g□◘9□|□◘=e\□up□i□|*P◆□□□Ow□ p□ko◆W(□□◘Xeb□[p□m□De□▼□QXD|□|►R[◘P[e□!□□D□l\U◘n*◘□|□XX□D□8/◆O□□Ve}□Vp}G□x◘□xe□I□|►T□x0□FH|□j\◘□|Q[◘□R[e□S[e□
□◘□ƨZ□&□◄e□<To@◆□□lp◄□□Q□wT|<|L|□P□w□%Wep□B►Ue◘|e□~▲□,h□9□B□□Y+□6□□□$[□w◄e◆AUOLe7/
```

Questions 7:



```
vol.py: error: unrecognized arguments: -- 5896
PS C:\Users\User\Documents\volatility-web\volatility3> python vol.py -f  C:\Users\User\Documents\volatility-web\memory\MemoryDump.mem windows.vadinfo --pid 5896
Volatility 3 Framework 2.28.2
Progress:  100.00               PDB scanning finished
PID     Process Offset  Start VPN       End VPN Tag     Protection      CommitCharge    PrivateMemory   Parent  File    File output

5896    oneetx.exe      0xffffad818ddb2c40      0x73da0000      0x74024fff      Vad     PAGE_EXECUTE_WRITECOPY  554     0       0x0     \Windows\SysWOW64\AcLayers.dll  Disabled
5896    oneetx.exe      0xffffad818ddc5200      0x1a60000       0x2e60fff       Vad     PAGE_READONLY   0       0       0xffffad818ddb2c40      N/A     Disabled
5896    oneetx.exe      0xffffad818d6d5b70      0x1390000       0x139ffff       Vads    PAGE_READWRITE  10      1       0xffffad818ddc5200      N/A     Disabled
5896    oneetx.exe      0xffffad818d6bc2b0      0x1f90000       0x1ffffff       Vads    PAGE_READWRITE  11      1       0xffffad818d6d5b70      N/A     Disabled
5896    oneetx.exe      0xffffad818ddab1c0      0xfc0000        0xfc0fff        Vad     PAGE_READONLY   0       0       0xffffad818d6bc2b0      N/A     Disabled
5896    oneetx.exe      0xffffad818d2c7ac0      0xec0000        0xfb7fff        Vad     PAGE_EXECUTE_WRITECOPY  0       0       0xffffad818ddab1c0      \Users\Tammam\AppData\Local\Temp\c3912
af058\oneetx.exe        Disabled
5896    oneetx.exe      0xffffad818d6d5940      0x400000        0x437fff        VadS    PAGE_EXECUTE_READWRITE  56      1       0xffffad818d2c7ac0      N/A     Disabled
5896    oneetx.exe      0xffffad818ddade20      0xfe0000        0xfe0fff        Vad     PAGE_READONLY   0       0       0xffffad818ddab1c0      N/A     Disabled
5896    oneetx.exe      0xffffad818ddab940      0xfd0000        0xfd0fff        Vad     PAGE_READONLY   0       0       0xffffad818ddade20      N/A     Disabled
5896    oneetx.exe      0xffffad81876f1b30      0xff0000        0xff3fff        Vad     PAGE_READONLY   0       0       0xffffad818ddade20      N/A     Disabled
5896    oneetx.exe      0xffffad818d6cb940      0x1360000       0x136ffff       VadS    PAGE_READWRITE  2       1       0xffffad818d6bc2b0      N/A     Disabled
5896    oneetx.exe      0xffffad818d6c5cc0      0x1220000       0x125ffff       VadS    PAGE_READWRITE  11      1       0xffffad818d6cb940      N/A     Disabled
5896    oneetx.exe      0xffffad818d2cfea0      0x1200000       0x121cfff       Vad     PAGE_READONLY   0       0       0xffffad818d6c5cc0      N/A     Disabled
5896    oneetx.exe      0xffffad818d6c8a10      0x1260000       0x135ffff       VadS    PAGE_READWRITE  6       1       0xffffad818d6c5cc0      N/A     Disabled
5896    oneetx.exe      0xffffad818ddac840      0x1370000       0x137Offf       Vad     PAGE_READONLY   0       0       0xffffad818d6cb940      N/A     Disabled
5896    oneetx.exe      0xffffad818ddae780      0x1380000       0x138Offf       Vad     PAGE_READONLY   0       0       0xffffad818ddac840      N/A     Disabled
5896    oneetx.exe      0xffffad818d6d5c60      0x1510000       0x160ffff       VadS    PAGE_READWRITE  218     1       0xffffad818d6d5b70      N/A     Disabled
```

8 Final Answers

1. oneetx.exe
2. Rundll32.exe
3. PAGE_EXECUTE_READWRITE
4. outline.exe
5. 77.91.124.20
6. http://77.91.124.20/store/games/js/index.php
7. C:\Users\Tammam\AppData\Local\Temp\c3912af058\oneetx.exe

9 Lessons Learned & Remediation

9.1 Key Takeaways

- **Process and network correlation:** Correlating process information from windows.pslist and windows.pstree with network activity from windows.netscan helps identify relationships between suspicious processes and network connections.
- **Do not rely on a single indicator:** tun2socks.exe initially appeared to be related to the VPN activity, but further investigation identified Outline.exe. Multiple artifacts should be correlated before reaching a conclusion.
- **VAD analysis:** windows.vadinfo can help identify suspicious memory regions and protections such as PAGE_EXECUTE_READWRITE.
- **Memory-resident artifacts:** Searching memory for IP addresses, URLs, and command-line artifacts can reveal additional evidence of attacker activity.
- **PowerShell filtering:** Select-String is useful for quickly filtering large Volatility outputs by PID, process name, IP address, or other keywords.

9.2 Remediation (if applicable to a real-world parallel)

1. **Monitor suspicious process execution:** Unusual executables running from temporary or user-writable directories should be monitored and investigated.
2. **Implement network monitoring:** Monitor unexpected outbound connections and investigate connections to suspicious external IP addresses.
3. **Restrict execution from temporary directories:** Apply application-control policies to prevent unauthorized executables from running from locations such as %TEMP% and user-writable directories.
4. **Enable endpoint logging:** Maintain detailed process creation, command-line, PowerShell, and network telemetry to support incident investigation.
5. **Use endpoint detection controls:** Deploy EDR/endpoint monitoring to detect suspicious process relationships, abnormal memory permissions, and potentially malicious network activity.

10 References & Appendix

10.1 References

- CyberDefenders – Redline Challenge — Challenge overview and forensic investigation scenario.
- Volatility 3 Documentation — Documentation for memory-forensics plugins and framework usage.

10.2 Appendix

The following supplementary materials can be included to support the investigation while keeping the main walkthrough concise:

- Relevant windows.pslist output
- windows.pstree process-tree output
- windows.netscan network-connection output
- windows.vadinfo output for the suspicious process
- Relevant Select-String search results
- Screenshots of key findings and commands
- Extracted URLs, IP addresses, PIDs, and process relationships
- Additional forensic artifacts or evidence used during the investigation

> **Note:** Full memory dumps and other large forensic artifacts should generally be kept separately rather than embedded directly in the report.

End of Writeup